DRUGS · ANTIBIOTICS: CEPHALOSPORINS AND CARBAPENEMS

Oral cephalosporins: cefalexin and cefuroxime

Cefalexin (first generation) and cefuroxime (second generation) are the oral cephalosporins an ENT doctor uses. They are alternatives to penicillins for patients whose penicillin allergy was a non-severe rash, and cefalexin is a reliable drug for staphylococcal and streptococcal skin and wound infection. Cefuroxime adds better Haemophilus and Moraxella cover. Neither covers Pseudomonas, anaerobes, MRSA or enterococci, and both are associated with C. difficile.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

CEFALEXIN AND CEFUROXIME

	SUMMARY
Class	Cephalosporins: cefalexin first generation, cefuroxime second generation
How they work	Beta-lactams that inhibit cell wall cross-linking; more stable to staphylococcal penicillinase than penicillin
Spectrum	Cefalexin: S. aureus (not MRSA), streptococci, some E. coli. Cefuroxime: adds H. influenzae and Moraxella, including beta-lactamase producers
ENT uses	Cellulitis and wound infection (cefalexin); acute otitis media and sinusitis in non-severe penicillin allergy (cefuroxime); surgical prophylaxis (cefuroxime or cefazolin intravenously)
Adult dose	Cefalexin 500 mg two or three times daily (up to 1 to 1.5 g three or four times daily); cefuroxime axetil 250 to 500 mg twice daily; intravenous cefuroxime 750 mg to 1.5 g three times daily
Child dose	Cefalexin 12.5 mg/kg twice daily (25 mg/kg twice daily for severe infection); cefuroxime 15 mg/kg twice daily (maximum 250 mg per dose)
Avoid	Previous immediate (anaphylactic) penicillin or cephalosporin allergy

	SUMMARY
Pregnancy	Safe

2 CHAPTER TWO How they work

Mechanism and handling.

2.1 Mechanism and generations

Cephalosporins share the beta-lactam ring of penicillins but have a six-membered dihydrothiazine ring that resists many beta-lactamases. Moving from the first to the third generation, Gram-negative activity increases and Gram-positive activity falls; cefalexin is the best oral anti-staphylococcal cephalosporin, cefuroxime the best balanced for the respiratory organisms.

2.2 Absorption and handling

Cefalexin is almost completely absorbed; cefuroxime axetil is a prodrug absorbed better with food. Both are excreted by the kidney: reduce the dose when renal function is poor.

3 CHAPTER THREE Using them in ENT

Indications and doses.

3.1 Penicillin allergy

Ask what actually happened. A **non-severe delayed rash** is not a contraindication to a cephalosporin; cross-reactivity is low (about 1% to 2%) and is driven mainly by similar side chains (cefalexin shares one with amoxicillin, cefuroxime does not). After **anaphylaxis, angioedema, or a severe cutaneous reaction** (Stevens-

Johnson syndrome, DRESS), avoid all beta-lactams and choose a macrolide, doxycycline or clindamycin.

3.2 Skin, wound and soft tissue

Cefalexin covers the staphylococci and streptococci of cellulitis, wound infection and infected epidermoid cysts, and is an option for prophylaxis after ear and facial plastic surgery where cover is wanted.

3.3 Surgical prophylaxis

Intravenous cefazolin or cefuroxime at induction (with metronidazole if the pharynx is opened) is standard for clean-contaminated head and neck surgery; prophylaxis should not run beyond 24 hours.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- Diarrhoea and **C. difficile**, among the highest risks of any antibiotic class.
- Hypersensitivity, including rash, serum sickness-like reactions (cefaclor especially) and anaphylaxis.
- Rarely a positive Coombs test with haemolysis, and interstitial nephritis.

4.2 Interactions

Mild potentiation of warfarin. Probenecid raises their levels.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The **generations** and how the spectrum shifts.
- Cross-reactivity with penicillin and which allergy histories rule them out.
- That no oral cephalosporin covers **Pseudomonas** or anaerobes.

5.2 Common pitfalls

COMMON PITFALLS

- Prescribing a cephalosporin after documented penicillin anaphylaxis.
- Using them for necrotising otitis externa or a dental abscess without anaerobic cover.
- Prolonged prophylaxis beyond 24 hours.

Written for this app from BNF and BNF for Children; NICE and specialist society guidance; Scott-Brown; Cummings. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.